System A - Midpoint Geometry

Primitive: a set $\mathbf{P}$ of points and a ternary relation $M \subseteq \mathbf{P}^3$. We read $M(A, B, C)$ as "B is the midpoint of A and C".

Axiom 1 (Uniqueness). $M(A, B, C) \wedge M(A, B', C) \implies B = B'$.

Axiom 2 (Symmetry). $M(A, B, C) \implies M(C, B, A)$.

Axiom 3 (Reflexivity). $M(A, A, A)$.

Axiom 4 (Existence). $\forall A, C \in \mathbf{P},\ \exists B \in \mathbf{P}\ M(A, B, C)$.

Axiom 5 (Cancellation). $M(A, B, A) \implies A = B$.

System A - Midpoint Geometry

Primitive: a set $\mathbf{P}$ of points and a ternary relation $M \subseteq \mathbf{P}^3$. We read $M(A, B, C)$ as "B is the midpoint of A and C".

Axiom 6 (Uniqueness). $M(A, B, C) \wedge M(A, B', C) \implies B = B'$.

Axiom 7 (Symmetry). $M(A, B, C) \implies M(C, B, A)$.

Axiom 8 (Reflexivity). $M(A, A, A)$.

Axiom 9 (Existence). $\forall A, C \in \mathbf{P},\ \exists B \in \mathbf{P}\ M(A, B, C)$.

Axiom 10 (Cancellation). $M(A, B, A) \implies A = B$.

System A - Midpoint Geometry

Primitive: a set $\mathbf{P}$ of points and a ternary relation $M \subseteq \mathbf{P}^3$. We read $M(A, B, C)$ as "B is the midpoint of A and C".

Axiom 11 (Uniqueness). $M(A, B, C) \wedge M(A, B', C) \implies B = B'$.

Axiom 12 (Symmetry). $M(A, B, C) \implies M(C, B, A)$.

Axiom 13 (Reflexivity). $M(A, A, A)$.

System B – Reflection Geometry

Primitive: *a set* $\mathbf{P}$ *and a function* $R : \mathbf{P} \times \mathbf{P} \to \mathbf{P}$. *We write* $R_B(A)$ *for the reflection of* A *across* B.

Axiom 14 (Involutivity). $R_B(R_B(A)) = A$.

Axiom 15 (Fixed point). $R_B(B) = B$.

Axiom 16 (Transitivity of reflections). $R_C(R_B(A)) = R_{R_C(B)}(R_C(A))$.

Axiom 17 (Existence of midpoint)**.** *For any A, C there exists a unique B such that $R_B(A) = C$.*

Axiom 18 (Density)**.** *If $A \neq B$, then $R_A(B) \neq B$ and the set $\{R_A^n(B) \mid n \in \mathbb{Z}\}$ has no least positive element (formally, there exists a point between A and B).*

Axiom 19 (Existence)**.** $\forall A, C \in \mathbf{P},\ \exists B \in \mathbf{P}\ M(A, B, C)$.

Axiom 20 (Cancellation)**.** $M(A, B, A) \implies A = B$.

Axiom 21 (Density of Midpoints)**.** $A \neq B \implies \exists C\ M(A, C, B)$.

$$\mathcal{B}(A, B, C) \stackrel{\text{def}}{=} \big(\exists X\,(M(A, X, B) \wedge M(X, B, C))\big) \vee (A = B \wedge B = C).$$

Axiom 22 (Density of Midpoints)**.** $A \neq B \implies \exists C\ M(A, C, B)$.

System B - Reflection Geometry

Primitive: a set $\mathbf{P}$ and a function $R : \mathbf{P} \times \mathbf{P} \to \mathbf{P}$. We write $R_B(A)$ for the reflection of A across B.

Axiom 23 (Involutivity)**.** $R_B(R_B(A)) = A$.

Axiom 24 (Fixed point)**.** $R_B(B) = B$.

Axiom 25 (Transitivity of reflections)**.** $R_C(R_B(A)) = R_{R_C(B)}(R_C(A))$.

Axiom 26 (Existence of midpoint)**.** *For any A, C there exists a unique B such that $R_B(A) = C$.*

Axiom 27 (Density)**.** *If $A \neq B$, then $R_A(B) \neq B$ and the set $\{R_A^n(B) \mid n \in \mathbb{Z}\}$ has no least positive element (formally, there exists a point between A and B).*

$$\mathcal{B}(A, B, C) \stackrel{\text{def}}{=} \big(\exists X\,(M(A, X, B) \wedge M(X, B, C))\big) \vee (A = B \wedge B = C).$$

System A - Midpoint Geometry

Primitive: a set $\mathbf{P}$ of points and a ternary relation $M \subseteq \mathbf{P}^3$. We read $M(A, B, C)$ as "B is the midpoint of A and C".

Axiom 28 (Uniqueness)**.** $M(A, B, C) \wedge M(A, B', C) \implies B = B'$.

Axiom 29 (Symmetry)**.** $M(A, B, C) \implies M(C, B, A)$.

Axiom 30 (Reflexivity)**.** $M(A, A, A)$.

System B – Reflection Geometry

Primitive: *a set* $\mathbf{P}$ *and a function* $R : \mathbf{P} \times \mathbf{P} \to \mathbf{P}$*. We write* $R_B(A)$ *for the reflection of* A *across* B*.*

Axiom 31 (Involutivity)**.** $R_B(R_B(A)) = A.$

Axiom 32 (Fixed point)**.** $R_B(B) = B.$

Axiom 33 (Transitivity of reflections)**.** $R_C(R_B(A)) = R_{R_C(B)}(R_C(A)).$

Axiom 34 (Existence of midpoint)**.** *For any* A, C *there exists a unique* B *such that* $R_B(A) = C$*.*

Axiom 35 (Density)**.** *If* $A \neq B$*, then* $R_A(B) \neq B$ *and the set* $\{R_A^n(B) \mid n \in \mathbb{Z}\}$ *has no least positive element (formally, there exists a point between* A *and* B*).*

Axiom 36 (Existence)**.** $\forall A, C \in \mathbf{P},\ \exists B \in \mathbf{P}\ M(A, B, C).$

Axiom 37 (Cancellation)**.** $M(A, B, A) \implies A = B.$

Axiom 38 (Density of Midpoints)**.** $A \neq B \implies \exists C\ M(A, C, B).$

$$\mathcal{B}(A, B, C) \overset{\text{def}}{=} \big(\exists X\, (M(A, X, B) \wedge M(X, B, C))\big) \vee (A = B \wedge B = C).$$

Axiom 39 (Density of Midpoints)**.** $A \neq B \implies \exists C\ M(A, C, B).$

System B - Reflection Geometry

Primitive: a set $\mathbf{P}$ and a function $R : \mathbf{P} \times \mathbf{P} \to \mathbf{P}$. We write $R_B(A)$ for the reflection of A across B.

Axiom 40 (Involutivity)**.** $R_B(R_B(A)) = A.$

Axiom 41 (Fixed point)**.** $R_B(B) = B.$

Axiom 42 (Transitivity of reflections)**.** $R_C(R_B(A)) = R_{R_C(B)}(R_C(A)).$

Axiom 43 (Existence of midpoint)**.** *For any* A, C *there exists a unique* B *such that* $R_B(A) = C$*.*

Axiom 44 (Density)**.** *If* $A \neq B$*, then* $R_A(B) \neq B$ *and the set* $\{R_A^n(B) \mid n \in \mathbb{Z}\}$ *has no least positive element (formally, there exists a point between* A *and* B*).*

$$\mathcal{B}(A, B, C) \overset{\text{def}}{=} \big(\exists X\, (M(A, X, B) \wedge M(X, B, C))\big) \vee (A = B \wedge B = C).$$